



OMRON Builds HPC Infrastructure for R&D on AWS, Uses Optimal Computing Resources, and Leads Innovative Technology Development

Learn how OMRON in manufacturing built an HPC environment on AWS, significantly improving research efficiency.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#)

5 minutes

to build an HPC environment



Hi, I can connect you with an AWS representative or answer questions you have on AWS.



Up to 128 CPU cores

for parallel processing



Need more info? Highlight any text to get an explanation generated with AWS generative AI.



[OMRON](#) manufactures four types of products: industrial automation, healthcare equipment, social systems, and electronic components. The company's research and development (R&D) department built a high performance computing (HPC) environment used for things such as machine learning (ML) and computer-aided engineering (CAE) analysis on Amazon Web Services (AWS). It achieved the construction of research environments in a short period of time and a reduction in analysis time.

Currently, the company is developing innovative technologies while making flexible use of AWS resources.



Opportunity | Revitalizing Computing Infrastructure for OMRON

Under its long-term vision, a project called "Shaping the Future 2030 (SF2030)" , OMRON aims to "continue to create innovation driven by social needs through automation to empower people."

In the midterm management plan for the fiscal years 2022–24, the company is working on transforming two things: its business and its organizational capabilities.

To achieve the goals of SF2030, OMRON needed to provide an R&D environment that met the needs of developers. However, in the existing on-premises environment, OMRON felt limited in procuring computational resources.

"In the sensing and healthcare fields, there is an increase in research on image recognition using ML and simulating human movements. Complex simulation models are also increasing in the fields of data and CAE analysis," says Manabu Tsuda from the business process platform design department, the technology and intellectual property headquarters at OMRON.

"These studies require large amounts of computational resources, but server procurement and environment construction take time, making it impossible to respond to requests from R&D departments in a timely manner. There was a shortage of computational resources during the busy season, and users were waiting for resources."



Therefore, the company decided to transform the current computing infrastructure of its R&D department from the ground up.

Solution | Using Security and Account Management Functions to Build a Secure Environment

The company wanted a cloud service that could flexibly scale computational resources and adopted AWS. The reason for the selection was because AWS has been widely adopted and provides security and account management features that could build a secure environment.

"Due to the nature in which R&D handles confidential information, high security is essential," says Tsuda. "The decisive factors for choosing AWS were that it is used by financial institutions and government agencies with strict security requirements and that there are plenty of features to protect confidential information."

In addition to the abundance of managed services and the high degree of freedom of development, another factor was that [OMRON SOFTWARE](#) (OSK), OMRON's software development company, began training its engineers on AWS in 2019, so they had accumulated knowledge on in-house production development.

After deciding to adopt AWS, a proof of concept was carried out from January to March 2022 to confirm construction procedures, flexibility, availability, and so on. The design policy was finalized from April to July 2022, and after constructing a production environment, it was released as the first step in October 2022 to two divisions: the department that researches ML and image recognition within the technology and intellectual property division, and the department that researches CAE and data analysis.

Depending on requirements, computational resources use various instance types of [Amazon Elastic Compute Cloud](#) (Amazon EC2). The existing R&D and HPC environments on AWS are connected using [AWS Direct Connect](#), a service that creates a dedicated network connection to AWS, to verify security.

OMRON configures a multi-account environment according to each research theme, such as co-creation with other companies, HPC computation, and co-creation with other companies, and uses [AWS Control Tower](#) for overall account control.



"If you manage accounts individually, there can be oversights and omissions with respect to security measures," says Hidemasa Ono of the cloud technology group, architecture development department, OSK core technology center. "By using the managed service AWS Control Tower, we were able to balance governance and agility."

Furthermore, the company developed a front screen using a Web GUI so that developers in the research and development department can allocate computational resources even if they are not familiar with AWS.

"To free our researchers from learning how to configure on AWS Management Console, we created a web front end where anyone can build an environment," says Ono.

Outcome | Reducing the HPC Environment Set-Up Time from Months to Minutes

By using AWS, the time to build an HPC environment has been greatly shortened, and it is now possible to respond quickly to researchers' requests.

"We were able to reduce the time it takes from procurement to construction from months to 5 minutes, and this has had a dramatic effect," says Tsuda. "I told the developers, 'Once you open the portal, it's a PC in five minutes,' and had them use it."

Computational resources can now be used without an upper limit, and it is now possible to select an instance with optimal specifications according to the research content. OMRON also expects that queuing and schedule adjustments during the busy season will be virtually removed, which will reduce user stress.

The ML and image recognition research department, which started using the AWS environment ahead of time, uses both AWS and on-premises environments. In parallel with using computational resources, the company is also building ML models using Amazon SageMaker. "By using the managed service [Amazon SageMaker](#), we can shorten the time from data collection to model construction, which will facilitate more efficient research," says Naohiro Tabata from the business process platform design department, technology and intellectual property division at OMRON.

Also, by setting up multiple instances and processing them in parallel, OMRON expects that the processing time will be shortened and the operational load will be reduced.

"If we set up a parallel processing pipeline, we can achieve a significant reduction in processing time," says Tabata. "Since processing proceeds automatically simply by submitting a job, the burden on developers is also reduced, and time can be spent on the content of the research."

The CAE and data analysis research department has also gained various benefits, such as shortening analysis time.

"As a result of using computational resources of up to 128 CPU cores in the optimization process, there were cases where the analysis period, which used to take several weeks, was shortened to about half a day," says Kazuki Sato of the technology and intellectual property division, digital design center at OMRON. "Waiting time during computation has been reduced, making it possible to accelerate research speed. In simulation, the benefits of being able to unify environments are also significant, and by handing over AWS machine images along with models developed by the research department to the technical department, there is no longer a need to worry about differences in simulation software versions."



Currently, the new research department is about to begin using AWS with OMRON SINIC X , a strategic base for creating near-future designs using cutting-edge technologies such as artificial intelligence and robotics. It is also going to use AWS in robotics development, which is being worked on in collaboration with external research institutes. In the future, the company also plans to roll out AWS to other development departments of the technology and intellectual property headquarters.

“We will increase the number of departments that use AWS while accumulating results from previous cases and deploying the results obtained throughout the technology and intellectual property division,” says Tsuda. “I would like to gradually expand the scope of use of AWS in the early days of SF2030 and make cloud use in R&D a culture common to the group.”

About OMRON

As a leading automation company, OMRON is engaged in a wide range of businesses including industrial automation centered on factory automation; electronic components such as switches, connectors, and sensors; social systems such as automatic railway ticket gates and power conditioners for solar power generation; and healthcare such as blood pressure monitors and body composition meters. It provides products and services in more than 130 countries and regions.

AWS Services Used

AWS Direct Connect

The AWS Direct Connect cloud service is the shortest path to your AWS resources. While in transit, your network traffic remains on the AWS global network and never touches the public internet. This reduces the chance of hitting bottlenecks or unexpected increases in latency.

[Learn more »](#)

AWS Control Tower

Use AWS Control Tower to set up and operate your multi-account AWS environment with prescriptive controls designed to accelerate your cloud journey.

[Learn more »](#)

Amazon SageMaker



Amazon SageMaker is a fully managed service that brings together a broad set of tools to enable high-performance, low-cost machine learning (ML) for any use case.

[Learn more »](#)

Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 750 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

[Learn more »](#)

Get Started

Organizations of all sizes across all industries are transforming their businesses and delivering on their missions every day using AWS. Contact our experts and start your own AWS journey today.

[Contact Sales](#)

Learn About AWS

- What Is AWS?
- What Is Cloud Computing?
- AWS Accessibility
- AWS Inclusion, Diversity & Equity
- What Is DevOps?
- What Is a Container?
- What Is a Data Lake?
- What is Artificial Intelligence (AI)?
- What is Generative AI?
- What is Machine Learning (ML)?
- AWS Cloud Security
- What's New
- Blogs

Resources for AWS

- Getting Started
- Training and Certification
- AWS Solutions Library
- Architecture Center
- Product and Technical FAQs
- Analyst Reports
- AWS Partners

Developers on AWS

- Developer Center
- SDKs & Tools
- .NET on AWS
- Python on AWS
- Java on AWS
- PHP on AWS
- JavaScript on AWS

Help

- Contact Us
- Get Expert Help
- File a Support Ticket
- AWS re:Post
- Knowledge Center
- AWS Support Overview
- Legal
- AWS Careers





Amazon is an Equal Opportunity Employer: *Minority / Women / Disability / Veteran / Gender Identity / Sexual Orientation / Age.*

Language

- | عربي
- | Bahasa Indonesia
- | Deutsch
- | English
- | Español
- | Français
- | Italiano
- | Português
- | Tiếng Việt
- | Türkçe
- | Русский
- | ไทย
- | 日本語
- | 한국어
- | 中文 (简体)
- | 中文 (繁體)

Privacy

- |
- Accessibility

- |
- Site Terms

- |
- Cookie Preferences

|

© 2024, Amazon Web Services, Inc. or its affiliates. All rights reserved.